

WELL COMPLETION REPORT (SYNTHETIC)

OIL-DRK-012 - Dirok-012

NWIS prototype document - not an Oil India Limited operational record

API Number	OIL-ASM-000140
Field / District	Dirok / Dibrugarh
Block	OALP-III-08
Basin	Assam-Arakan
Formation	Tipam (TIP)
Spud / Completion	2003-03-26 / 2003-05-10
Measured / TVD	2496 m / 2459 m
Trajectory	Vertical; inc 4.5 deg; az 149.4 deg
Formation Interval	1145-1898 m
Lithology	Fine-medium sandstone with shale interbeds
Pore Pressure Reference	2435 psi
Fracture Gradient	14.73 ppg
Mud System	Low-Solids WBM
NWIS Well Risk	Moderate (43.0/100)

Casing Program

String	Set Depth (m)	Purpose
20 in conductor	80	Conductor
13 3/8 in casing	624	Surface
9 5/8 in casing	1447	Intermediate
7 in liner/casing	2196	Production

Significant Historical Events and Lessons

Date / Depth	Event	Severity	Evidence / mitigation / lesson
2003-04-26 1814 m	Kick	Critical	Flow-out exceeded flow-in and gas units increased. Mitigation: Shut in, verify influx and circulate using approved well-control procedure. Lesson: Treat pit gain plus gas increase as an early influx indicator.
2003-05-01 2025 m	Kick	Critical	Drilling break followed by positive flow check. Mitigation: Maintain trip margin and use conservative connection practices. Lesson: Maintain mud-weight window using offset pressure evidence.
2003-04-09 905 m	Mud Loss	High	Active pit volume declined during stable pump rate. Mitigation: Reduce ECD, treat losses with LCM and confirm stable returns. Lesson: Prepare LCM package before approaching correlated offset-well loss depth.
2003-04-13 1128 m	Torque Spike	High	Sharp torque excursion occurred with RPM instability. Mitigation: Ream and condition hole if torque remains elevated. Lesson: Torque-rate change is a useful precursor even when pressure is stable.
2003-04-17 1347 m	Pressure Spike	High	Short-duration high-pressure excursion was recorded. Mitigation: Condition hole and confirm pressure normalization. Lesson: Correlate pressure anomalies with flow before escalating risk.
2003-04-24 1679 m	Pressure Spike	High	Short-duration high-pressure excursion was recorded. Mitigation: Reduce pump rate, verify restriction and restore circulation gradually. Lesson: Rate-of-change in pressure is more informative than a fixed threshold.
2003-04-27 1834 m	Kick	High	Flow-out exceeded flow-in and gas units increased. Mitigation: Maintain trip margin and use conservative connection practices. Lesson: Treat pit gain plus gas increase as an early influx indicator.

Completion Summary

The synthetic well reached 2496 m MD in the Tipam interval. The NWIS knowledge record links 12 historical operational events to depth-matched telemetry. Risk classification is derived from formation, depth, trajectory and event history rather than formation alone.

This report is generated solely for SIH/NWIS prototyping and model development.